

Standardizing Hybrid Automata: From
Theoretical Foundations to Real-World
Implementation Frameworks

A Pilot Literature on the Development of a Unified Framework for
Hybrid Automata Development and Simulation

Ryan McKee

Feb 2026

Abstract

Hybrid automata are formal models for a dynamical system with discrete and continuous components [?] widely adopted across research & industry for a variety of use cases ranging from control systems to stock trading algorithms, and from robotics to cyber-physical systems a recent example of this is for solving COLREG compliant autonomy for maritime vehicles [?] where the hybrid automata is used to model the discrete decision making process of the vessel and the continuous dynamics of the vessel's movement.

This theoretical framework provides a powerful method for modeling complex systems that can solve real world problems algorithmically without the need for human intervention during the execution of the system in a way that is efficient and effective and totally transparent for the human-in-the-loop to comprehend the decision making process of the system.

While in recent years there has been significant progress in the development of Artificial Intelligence (AI) through the rise of large-scale models like OpenAI's GPT series [?] that have shown the ability to imitate human-like reasoning further blurring the lines between human and machine intelligence therefore reducing the gap between what humans and machines can do there are still issues with AI that make it difficult to deploy in real world application such as lack of transparency and explainability due to the black-box nature of decision making caused by the topology of deep-neural networks (dnn) and also an issue with the reliability of AI systems due to their sensitivity to mistakes in novel situations that they have not been trained on and undefined emergent behaviours arising from outliers in their training data.

for the reasons presented above, hybrid automaton is still one of the most powerful tools for complex system design however, the theoretical framework has been laid there seems to be a lack of standardization in converting the theoretical framework into real world applications with several papers implementing their own versions of hybrid automaton with different algorithms enabling them.

In this pilot literature review, we will explore the current state of the art in hybrid automata development and simulation to identify the current gaps in the field and propose a unified framework for hybrid automata development and simulation that can be adopted across research and industry to standardize the development process and enable more efficient and effective use of hybrid automata in real-world applications.

0.1 introduction

Hybrid automata are formal models for a dynamical system with discrete and continuous components [?] widely adopted across research & industry for a variety of use cases ranging from control systems to stock trading algorithms, and from robotics to cyber-physical systems a recent example of this is for solving COLREG compliant autonomy for maritime vehicles [?] where the hybrid automata is used to model the discrete decision making process of the vessel and the continuous dynamics of the vessel's movement.

This theoretical framework provides a powerful method for modeling complex systems that can solve real world problems algorithmically without the need for human intervention during the execution of the system in a way that is efficient and effective and totally transparent for the human-in-the-loop to comprehend the decision making process of the system.

While in recent years there has been significant progress in the development of Artificial Intelligence (AI) through the rise of large-scale models like OpenAI's GPT series [?] that have shown the ability to imitate human-like reasoning further blurring the lines between human and machine intelligence therefore reducing the gap between what humans and machines can do there is still issues with AI that make it difficult to deploy in real world application such as lack of transparency and explainability due to the black-box nature of decision making caused by the topology of deep-neural networks (dnn) and also an issue with the reliability of AI systems due to their sensitivity to mistakes in novel situations that they have not been trained on and undefined emergent behaviours arising from outliers in their training data.

for the reasons presented above, hybrid automaton is still one of the most powerful tools for complex system design however, the theoretical framework has been laid there seems to be a lack of standardization in converting the theoretical framework into real world applications with several papers implementing their own versions of hybrid automaton with different algorithms enabling them.

In this pilot literature review, we will explore the current state of the art in hybrid automata development and simulation to identify the current gaps in the field and propose a unified framework for hybrid automata development and simulation that can be adopted across research and industry to standardize the development process and enable more efficient and effective use of hybrid automata in real-world applications.

0.2 background

Hybrid automata are extensions of finite automata therefore if you are familiar with them you will notice many similarities between the two, however, the key difference being the incorporation of continuous variables and dynamics as well

as resets on transitions that effect auxiliary states.

The syntax for modelling a hybrid automaton is as follows...

$$H = (Q, X, f, Init, Inv, E, G, R)$$

where:

- Q : a finite set of discrete states
- X : a finite set of continuous variables
- f : a function defining the continuous dynamics in each discrete state
- $Init$: a set of initial conditions
- Inv : a set of invariants for each discrete state
- E : a set of edges defining transitions
- G : guards enabling transitions
- R : reset maps updating continuous variables during transitions

Figure 1: Syntax for modelling a hybrid automaton

A simple example for reference is a thermostate system, to show the above syntax way define the definition of a thermostat system utilizing hybrid automaton as follows:

$$H = (Q, X, f, Init, Inv, E, G, R)$$

where:

- $Q = \{ON, OFF\}$: the discrete states representing the thermostat being on or off
- $X = \{T\}$: the continuous variable representing the temperature
- f : the dynamics defined as $f(ON) = -k(T - T_{set})$ and $f(OFF) = k(T_{ambient} - T)$ where k is a constant and T_{set} and $T_{ambient}$ are the set-point and ambient temperatures respectively
- $Init$: the initial conditions could be defined as $Init = \{(OFF, T_0)\}$ where T_0 is the initial temperature
- Inv : the invariants could be defined as $Inv(ON) = \{T < T_{set}\}$ and $Inv(OFF) = \{T > T_{ambient}\}$
- E : the edges defining transitions could be defined as $E = \{(OFF, ON), (ON, OFF)\}$
- G : the guards enabling transitions could be defined as $G(OFF, ON) = \{T < T_{set}\}$ and $G(ON, OFF) = \{T > T_{ambient}\}$
- R : the reset maps updating continuous variables during transitions could be defined as $R(OFF, ON) = T$ and $R(ON, OFF) = T$

Figure 2: Example of a hybrid automaton for a thermostat system

0.3 State of the art

0.4 Discussion / Reserach Gaps

Currently there is alot of work on automaton, but there is a lack of stardaization in bridgin the gap between the theoricial definition and the actual implemen-tation of the autoamton in the real world. Projects like [?] however they arent really focused on stardization of development of hybrid automaton and are quite difficult to use, Therefore there is a need for a unified framework for hybrid automata development and simulation that can be adopted across research and industry to standardize the development process and enable more efficient and effective use of hybrid automata in real-world applications.

0.5 conclusions / Future Work

The gap in research/industry prevented outlines the need for a unified frame-owork for automaton Development, therefore I propose the development of a unified framework and definition langauge for programatically converting au-tomaton to real world application that can be adopted across research and industry. I propose that initially to make this a simple frameowrk anyone can use to create a python-based framework for simulating and running automata in real-time and offline settings. It should provide a lightweight, flexible foun-dation for defining evaluating custom automata whether that be for research of real world application, whether is be a simple thermostate system or an au-tonomy stack. It should not only be able to simulate the automaton but also provide tools for visualizing the automaton like in figureand also provide tools for evaluating the automaton in real world settings with real world data.

Bibliography

- [1] T.A. Henzinger. The theory of hybrid automata. In *Proceedings 11th Annual IEEE Symposium on Logic in Computer Science*, pages 278–292, 1996.
- [2] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, Red Avila, Igor Babuschkin, Suchir Balaji, Valerie Balcom, Paul Baltescu, Haiming Bao, Mohammad Bavarian, Jeff Belgum, Irwan Bello, Jake Berdine, Gabriel Bernadett-Shapiro, Christopher Berner, Lenny Bogdonoff, Oleg Boiko, Madelaine Boyd, Anna-Luisa Brakman, Greg Brockman, Tim Brooks, Miles Brundage, Kevin Button, Trevor Cai, Rosie Campbell, Andrew Cann, Brittany Carey, Chelsea Carlson, Rory Carmichael, Brooke Chan, Che Chang, Fotis Chantzis, Derek Chen, Sully Chen, Ruby Chen, Jason Chen, Mark Chen, Ben Chess, Chester Cho, Casey Chu, Hyung Won Chung, Dave Cummings, Jeremiah Currier, Yunxing Dai, Cory Decareaux, Thomas Degry, Noah Deutsch, Damien Deville, Arka Dhar, David Dohan, Steve Dowling, Sheila Dunning, Adrien Ecoffet, Atty Eleti, Tyna Eloundou, David Farhi, Liam Fedus, Niko Felix, Simón Posada Fishman, Juston Forte, Isabella Fulford, Leo Gao, Elie Georges, Christian Gibson, Vik Goel, Tarun Gogineni, Gabriel Goh, Rapha Gontijo-Lopes, Jonathan Gordon, Morgan Grafstein, Scott Gray, Ryan Greene, Joshua Gross, Shixiang Shane Gu, Yufei Guo, Chris Hallacy, Jesse Han, Jeff Harris, Yuchen He, Mike Heaton, Johannes Heidecke, Chris Hesse, Alan Hickey, Wade Hickey, Peter Hoeschele, Brandon Houghton, Kenny Hsu, Shengli Hu, Xin Hu, Joost Huizinga, Shantanu Jain, Shawn Jain, Joanne Jang, Angela Jiang, Roger Jiang, Haozhun Jin, Denny Jin, Shino Jomoto, Billie Jonn, Heewoo Jun, Tomer Kaftan, Łukasz Kaiser, Ali Kamali, Ingmar Kanitscheider, Nitish Shirish Keskar, Tabarak Khan, Logan Kilpatrick, Jong Wook Kim, Christina Kim, Yongjik Kim, Jan Hendrik Kirchner, Jamie Kiros, Matt Knight, Daniel Kokotajlo, Łukasz Kondraciuk, Andrew Kondrich, Aris Konstantinidis, Kyle Kosic, Gretchen Krueger, Vishal Kuo, Michael Lampe, Ikai Lan, Teddy Lee, Jan Leike, Jade Leung, Daniel Levy, Chak Ming Li, Rachel Lim, Molly Lin, Stephanie Lin, Mateusz Litwin, Theresa Lopez, Ryan Lowe, Patricia Lue, Anna Makanju, Kim Malfacini, Sam Manning, Todor Markov, Yaniv Markovski, Bianca Martin, Katie Mayer, Andrew Mayne, Bob Mc-

Grew, Scott Mayer McKinney, Christine McLeavey, Paul McMillan, Jake McNeil, David Medina, Aalok Mehta, Jacob Menick, Luke Metz, Andrey Mishchenko, Pamela Mishkin, Vinnie Monaco, Evan Morikawa, Daniel Mossing, Tong Mu, Mira Murati, Oleg Murk, David Mély, Ashvin Nair, Reiichiro Nakano, Rameev Nayak, Arvind Neelakantan, Richard Ngo, Hyeonwoo Noh, Long Ouyang, Cullen O’Keefe, Jakub Pachocki, Alex Paino, Joe Palermo, Ashley Pantuliano, Giambattista Parascandolo, Joel Parish, Emy Parparita, Alex Passos, Mikhail Pavlov, Andrew Peng, Adam Perelman, Filipe de Avila Belbute Peres, Michael Petrov, Henrique Ponde de Oliveira Pinto, Michael, Pokorny, Michelle Pokrass, Vitchyr H. Pong, Tolly Powell, Alethea Power, Boris Power, Elizabeth Proehl, Raul Puri, Alec Radford, Jack Rae, Aditya Ramesh, Cameron Raymond, Francis Real, Kendra Rimbach, Carl Ross, Bob Rotsted, Henri Roussez, Nick Ryder, Mario Saltarelli, Ted Sanders, Shibani Santurkar, Girish Sastry, Heather Schmidt, David Schnurr, John Schulman, Daniel Selsam, Kyla Sheppard, Toki Sherbakov, Jessica Shieh, Sarah Shoker, Pranav Shyam, Szymon Sidor, Eric Sigler, Maddie Simens, Jordan Sitkin, Katarina Slama, Ian Sohl, Benjamin Sokolowsky, Yang Song, Natalie Staudacher, Felipe Petroski Such, Natalie Summers, Ilya Sutskever, Jie Tang, Nikolas Tezak, Madeleine B. Thompson, Phil Tillet, Amin Tootoonchian, Elizabeth Tseng, Preston Tuggle, Nick Turley, Jerry Tworek, Juan Felipe Cerón Uribe, Andrea Vallone, Arun Vijayvergiya, Chelsea Voss, Carroll Wainwright, Justin Jay Wang, Alvin Wang, Ben Wang, Jonathan Ward, Jason Wei, CJ Weinmann, Akila Welihinda, Peter Welinder, Jiayi Weng, Lilian Weng, Matt Wiethoff, Dave Willner, Clemens Winter, Samuel Wolrich, Hannah Wong, Lauren Workman, Sherwin Wu, Jeff Wu, Michael Wu, Kai Xiao, Tao Xu, Sarah Yoo, Kevin Yu, Qiming Yuan, Wojciech Zaremba, Rowan Zellers, Chong Zhang, Marvin Zhang, Shengjia Zhao, Tianhao Zheng, Juntang Zhuang, William Zhuk, and Barret Zoph. Gpt-4 technical report, 2024.

- [3] Bhawana Singh, Karim Ahmadi Dastgerdi, Nikolaos Athanasopoulos, Wasif Naeem, and Benoit Lecallard. Safe colregs-aware finite-time guidance and control of marine vehicles. *Ocean Engineering*, 351:123919, 2026.